

Tutorial 8

Monday:

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Question 2

Problem 61

Monday:

Question 4

The definition of the impulse on an object is

$$\Delta \vec{p} = \vec{F} \Delta t$$

or

$$d\vec{p} = \vec{F} dt \therefore \int d\vec{p} = \int \vec{F} dt$$

which is the area under the F vs t curve.

Using this we can determine the area under the curves and compare these areas.

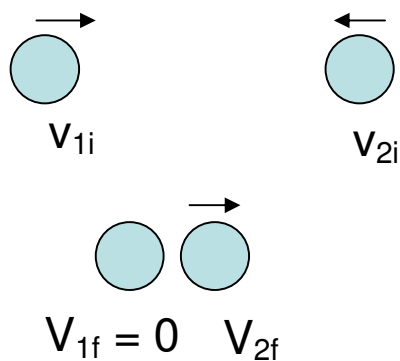
a) Area = $2F_0 \times 6t_0 = 12 F_0 t_0$

b) Area = $4F_0 \times 3t_0 = 12 F_0 t_0$

c) Area = $\frac{1}{2} (2F_0 \times 12t_0) = 12 F_0 t_0$

In this case all the areas are the same ✓✓✓

Problem 62



(a) The student must first define which particle is the one which is stationary before the collision. They can either use the conventional formula for conservation of momentum and conservation on Kinetic energy to solve the equation, or they can use the formula given in the textbook (Eq 9-75 given below) v_{1i} and v_{2i} must have same magnitude. Divide the marks according to the method used. The students have their textbooks and therefore can use the formula Eq 9-75.

Let m_1 be the mass of one sphere, v_{1i} be its velocity before the collision, and v_{1f} be its velocity after the collision. ✓

Let m_2 be the mass of the other sphere, v_{2i} be its velocity before the collision, and v_{2f} be its velocity after the collision. ✓ Then, according to Eq. 9-75,

$$v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_{1i} + \frac{2m_2}{m_1 + m_2} v_{2i}. \quad (\text{Eq 9-75})$$

Suppose sphere 1 is originally traveling in the positive direction and is at rest after the collision. ✓ Sphere 2 is originally traveling in the negative direction. ✓ Replace v_{1i} with v , v_{2i} with $-v$ (they are moving in opposite directions), and v_{1f} with zero to obtain $0 = m_1 - 3m_2$. Thus,

$$m_2 = m_1 / 3 = (250 \text{ g}) / 3 = 83.3 \text{ g} \checkmark$$

(b) We use the velocities before the collision to compute the velocity of the center of mass:

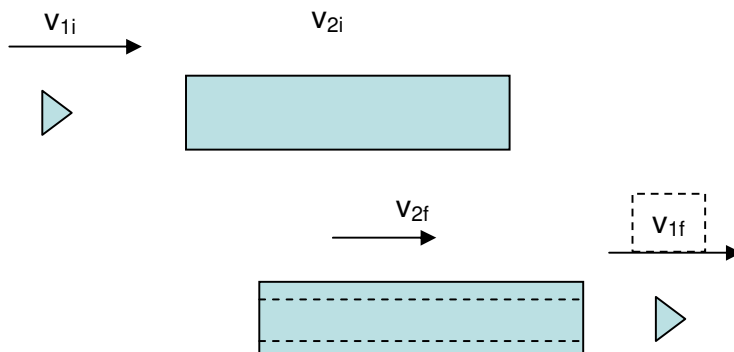
$$v_{com} = \frac{m_1 v_{1i} + m_2 v_{2i}}{m_1 + m_2} = \frac{(250 \text{ g})(2.00 \text{ m/s}) + (83.3 \text{ g})(-2.00 \text{ m/s})}{(250 \text{ g} + 83.3 \text{ g})} = 1.00 \text{ m/s} \checkmark \checkmark$$

Tuesday:

Question 2

- a) particles (c) and (d) and particles (c) and (b) have stationary COMs $\checkmark \checkmark$
- b) particles (c) and (b) will have stationary COM at the origin. $\checkmark$
- c) Particles (b) and (d) COM will pass through the origin $\checkmark$. Particles (a) and (d) COM will initially be at the origin, but will then move off the origin.

Problem 50



(a) We choose $+x$ along the initial direction of motion and apply momentum conservation:

$$m_{\text{bullet}} \vec{v}_i = m_{\text{bullet}} \vec{v}_1 + m_{\text{block}} \vec{v}_2 \checkmark \checkmark \checkmark$$

$$(5.2)(700) = (5.2)(450) + (700)v_2$$

which yields $v_2 = 1.86 \text{ m/s}$.

(b) It is a consequence of momentum conservation that the velocity of the center of mass is unchanged by the collision. We choose to evaluate it before the collision:

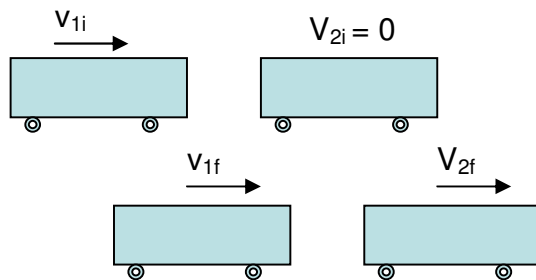
$$\vec{v}_{\text{com}} = \frac{m_{\text{bullet}} \vec{v}_i}{m_{\text{bullet}} + m_{\text{block}}} = \frac{(5.2 \text{ g})(700 \text{ m/s})}{5.2 \text{ g} + 700 \text{ g}} = 5.16 \text{ m/s. } \checkmark \checkmark \checkmark$$

Wednesday (Thursday 2 May)

Question 2

- a) particles (c) and (d) and particles (c) and (b) have stationary COMs $\checkmark \checkmark$
- b) particles (c) and (b) will have stationary COM at the origin. $\checkmark$
- c) Particles (b) and (d) COM will pass through the origin $\checkmark$. Particles (a) and (d) COM will initially be at the origin, but will then move off the origin.

Problem 61



Let m_1 be the mass of the cart that is originally moving, v_{1i} be its velocity before the collision, and v_{1f} be its velocity after the collision.

Let m_2 be the mass of the cart that is originally at rest and v_{2f} be its velocity after the collision.

Conservation of linear momentum gives $m_1 v_{1i} = m_1 v_{1f} + m_2 v_{2f}$.

Similarly, the total kinetic energy is conserved and we have $\frac{1}{2} m_1 v_{1i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2$.

Solving for v_{1f} and v_{2f} , we obtain (These equations are in the text book and can be used as is or they can be derived from the two equations above for momentum and KE):

$$v_{1f} = \frac{m_1 - m_2}{m_1 + m_2} v_{1i}, \quad v_{2f} = \frac{2m_1}{m_1 + m_2} v_{1i}$$

The speed of the center of mass is $v_{\text{com}} = \frac{m_1 v_{1i} + m_2 v_{2i}}{m_1 + m_2}$.

(a) With $m_1 = 0.34 \text{ kg}$, $v_{1i} = 1.2 \text{ m/s}$ and $v_{1f} = 0.66 \text{ m/s}$, we obtain

$$m_2 = \frac{v_{1i} - v_{1f}}{v_{1i} + v_{1f}} m_1 = \left(\frac{1.2 \text{ m/s} - 0.66 \text{ m/s}}{1.2 \text{ m/s} + 0.66 \text{ m/s}} \right) (0.34 \text{ kg}) = 0.0987 \text{ kg} \approx 0.099 \text{ kg}. \checkmark \checkmark$$

(b) The velocity of the second cart is:

$$v_{2f} = \frac{2m_1}{m_1 + m_2} v_{1i} = \left(\frac{2(0.34 \text{ kg})}{0.34 \text{ kg} + 0.099 \text{ kg}} \right) (1.2 \text{ m/s}) = 1.9 \text{ m/s}. \checkmark \checkmark$$

(c) From the above, we find the speed of the center of mass to be

$$v_{\text{com}} = \frac{m_1 v_{1i} + m_2 v_{2i}}{m_1 + m_2} = \frac{(0.34 \text{ kg})(1.2 \text{ m/s}) + 0}{0.34 \text{ kg} + 0.099 \text{ kg}} = 0.93 \text{ m/s}. \checkmark \checkmark$$

Note: In solving for v_{com} , values for the initial velocities were used. Since the system is isolated with no external force acting on it, v_{com} remains the same after the collision, so the same result is obtained if values for the final velocities are used. That is,

$$v_{\text{com}} = \frac{m_1 v_{1f} + m_2 v_{2f}}{m_1 + m_2} = \frac{(0.34 \text{ kg})(0.66 \text{ m/s}) + (0.099 \text{ kg})(1.9 \text{ m/s})}{0.34 \text{ kg} + 0.099 \text{ kg}} = 0.93 \text{ m/s}.$$